

Econ 101A

Section 10

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1 Production Functions

1.1 Definition and Description

- A **production function** $f : \mathbb{R}_+^n \rightarrow \mathbb{R}_+$ maps *inputs* (factors of production) to *outputs* (goods and services).
- Usually, we consider the two input case where some good $y = f(K, L)$ is produced using capital K and labor L .
- Common assumptions about production functions:
 - $f(\mathbf{0}) = 0$ (“no free lunch” - if you have no inputs, you will produce nothing)
 - $\frac{\partial f}{\partial z_i} > 0$ for all i (positive marginal productivity for all inputs z_i)
 - $\frac{\partial^2 f}{\partial z_i^2} < 0$ for all i (decreasing marginal productivity for all inputs z_i)
 - (in two-input case) $\frac{d^2 K}{dL^2} > 0$ (convex isoquants, which implies a convex production function)¹

1.2 Isoquants, two variable case (compare to indifference curves!)

- An **isoquant** is a curve representing different combinations of inputs that yield the same amount of output, or the line represented by the equation $f(K, L) = y_0$ for the output level y_0 .
- The **marginal rate of technical substitution (MRTS)** is the rate at which we substitute out capital for an additional unit of labor to remain at the same level of production.
- MRTS is also the absolute value of the slope of the isoquant. Assuming positive marginal productivity, and applying the Implicit Function Theorem, we get

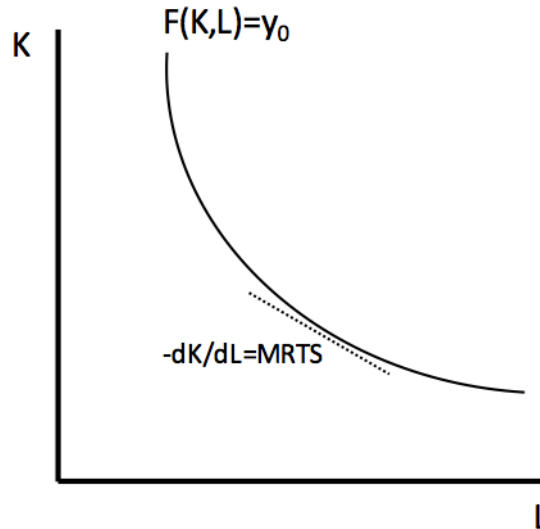
$$MRTS = -\frac{dK}{dL} = \frac{\partial f / \partial L}{\partial f / \partial K}$$

- Below is an example isoquant graph (assuming all assumptions given in Section 1.1). often graph isoquants with capital K as the vertical axis and labor L as the horizontal axis. Notice the following, which are similar to indifference curves:
 - As we move northeast (on the graph), we reach isoquants that indicate higher levels of production

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¹Note that a sufficient condition for this to hold is that $\frac{\partial^2 f}{\partial K \partial L} > 0$ - this says that the marginal productivity of labor increases with more capital and vice versa.

- Our isoquants are convex, indicating that a mixture of capital and labor will allow us to produce more than using *only* capital or *only* labor.



- Note: *unlike* with utility, you *cannot* do monotonic transformations of a production function and expect the same result.
 - Recall that in utility maximization problems without uncertainty, the utility function $U(x,y) = (x+y)^{1/2}$ would lead to the same choices as $U(x,y) = (x+y)$ or $U(x,y) = (x+y)^2$. These functions represent the same preferences (excluding risk preferences), since each function is a monotonic transformation of the others.
 - In utility maximization problems, only the *nominal* level of U would be affected, and this had no interpretable meaning - we only care about the *order* when thinking about preferences.
 - In production problems, however, Y has real meaning: it is the quantity produced.
 - Thus, $f(L,K) = (L+K)^{1/2}$ differs meaningfully from $f(L,K) = (L+K)$ or $f(L,K) = (L+K)^2$. Each will lead to very different production decisions (notice they have different returns to scale).

1.3 Returns to Scale

- f has **constant returns to scale** if for all $\mathbf{z}$ and for all $t > 1$, $f(t\mathbf{z}) = tf(\mathbf{z})$
- f has **decreasing returns to scale** if for all $\mathbf{z}$ and for all $t > 1$, $f(t\mathbf{z}) < tf(\mathbf{z})$
- f has **increasing returns to scale** if for all $\mathbf{z}$ and for all $t > 1$, $f(t\mathbf{z}) > tf(\mathbf{z})$

Given a production function $f(k,l)$, how do we determine whether it has increasing, decreasing, or constant returns to scale?

1. Obtain $f(tK,tL)$ by plugging in tK for every K and tL for every L that shows up in $f(K,L)$.
2. Rearrange terms so that you can easily compare your answer to $tf(K,L)$. When comparing, remember that $t > 1$.

2 Exercises

2.1 Isoquants

Draw one isoquant for each of the following production functions:

- $y = K^\alpha L^{1-\alpha}$, where $0 < \alpha < 1$
- $y = K + L$
- $y = \min\{K, L\}$

In a new graph, draw the isoquants of two production processes that exhibits (a) constant returns to scale, and (b) increasing returns to scale.

2.2 Returns to Scale

Determine whether the following production processes exhibit decreasing, constant, or increasing returns to scale.

- (a) Special Case Perfect Substitutes: $y = (L + K)^{1/2}$
- (b) General Cobb-Douglas: $f(L, K) = L^\alpha K^\beta$

2.3 Returns to Scale (Agriculture)

Let $f(L, K, F) = AL^\alpha K^\beta F^\gamma$ where F is farmland.

1. Show that $f(L, K, F)$ exhibits constant returns to scale if $\alpha + \beta + \gamma = 1$. Interpret this result.
2. Consider a small farm with the production function written above, with $\alpha + \beta + \gamma = 1$. In the short-run, however, the farm owner owns a fixed amount of land $F = \bar{F}$. Thus, his short-run production function is $Y_{SR} = AL^\alpha K^\beta \bar{F}^\gamma$. Does this production function have increasing, decreasing, or constant returns to scale in L and K ?